Turret Main Doc

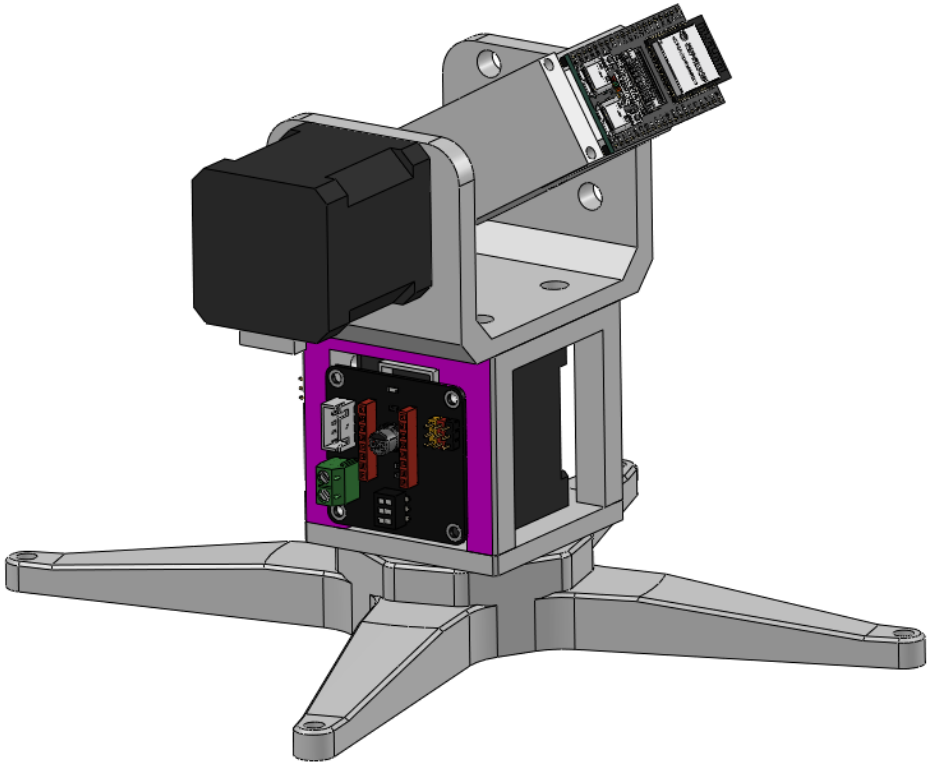
Summer 2025

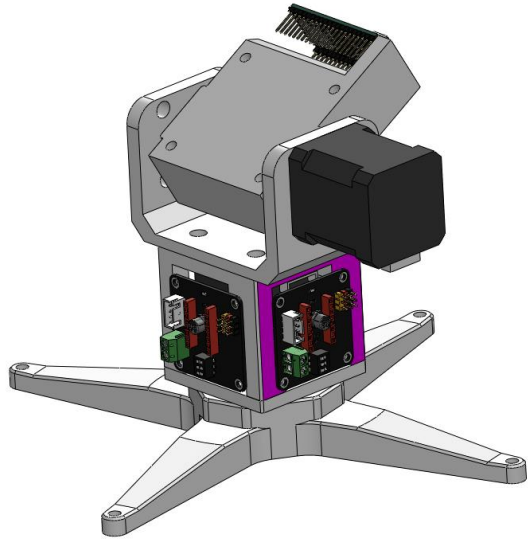
Brainstorming

- 2 axis
- Gas or Spring for foam darts
- Motors? Steppers or BLDC
- Controls? ESP32 or Raspberry Pi
- Does base motor spin the top or itself.

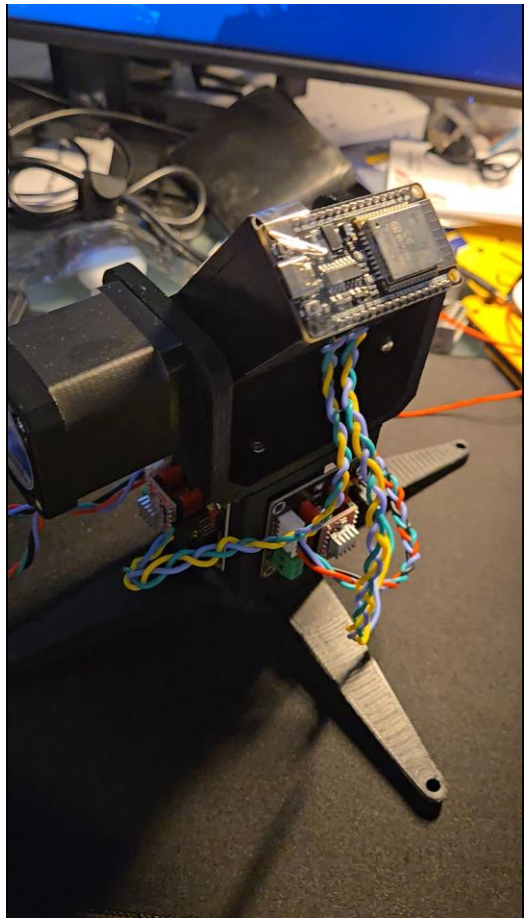
- Gas, Stepper, ESP32, coupled.
- Gas and stepper because cheap and easy.
- ESP 32 because cheap.
- Itself because we wanted 360 without wire tangling

First Prototype



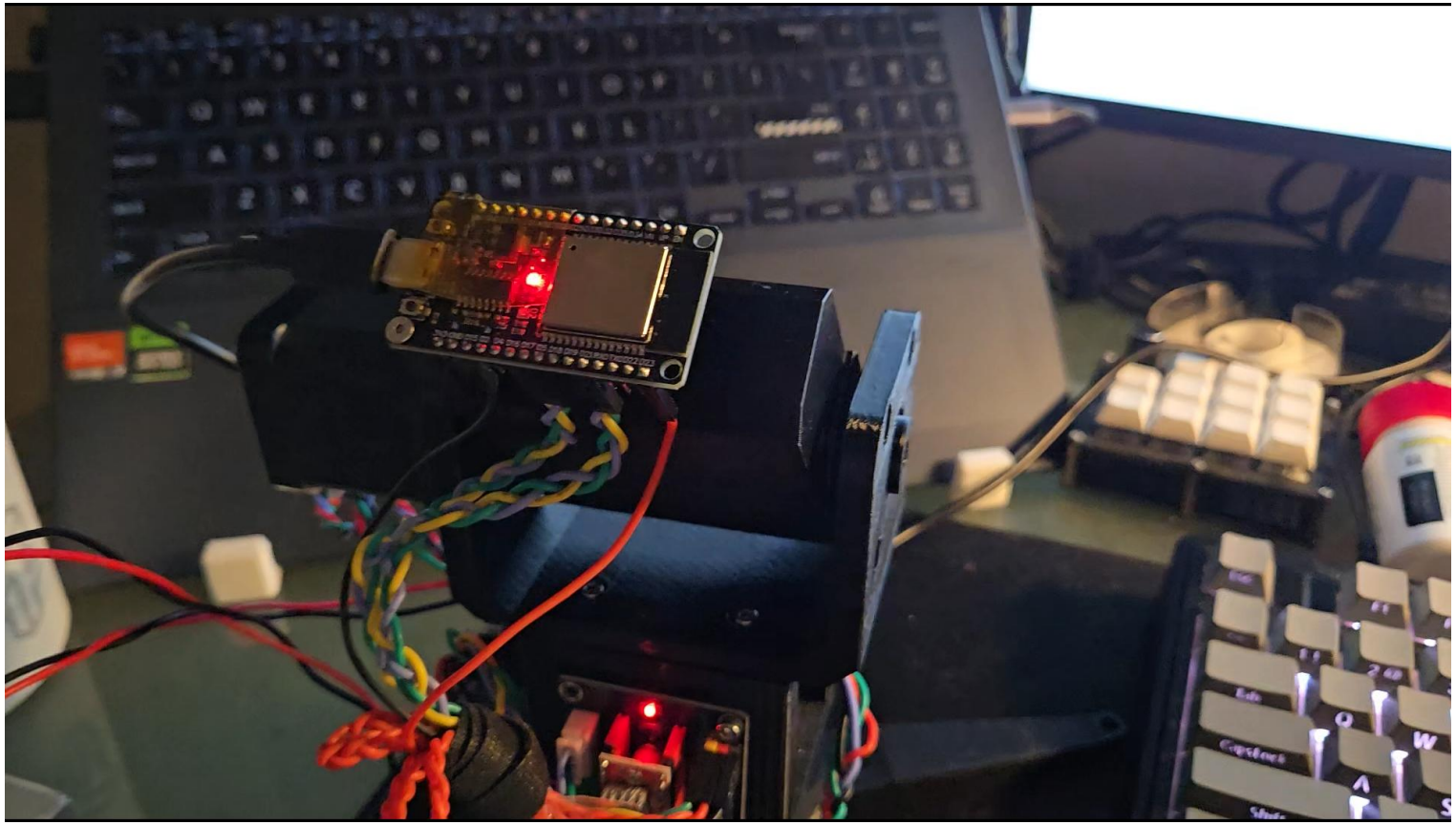


Spins freely.



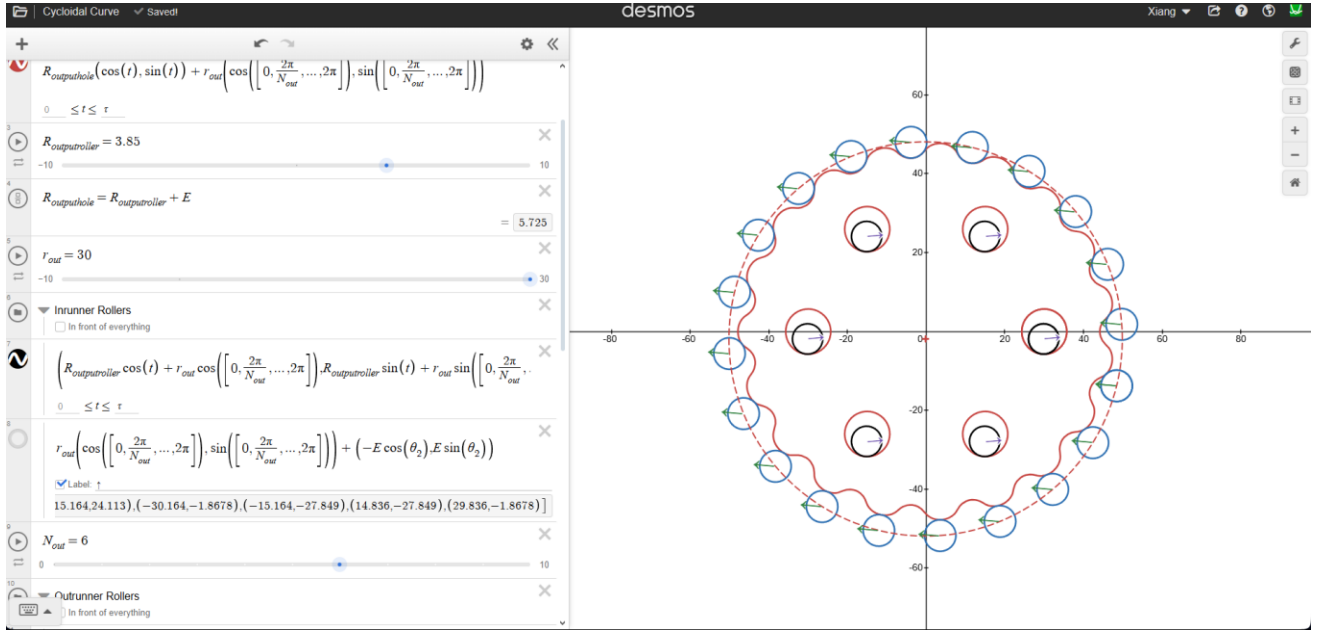
But the whole top part was very heavily and the motor could not move it.

Issues



But the whole top part was very heavily and the bottom motor could not move it.

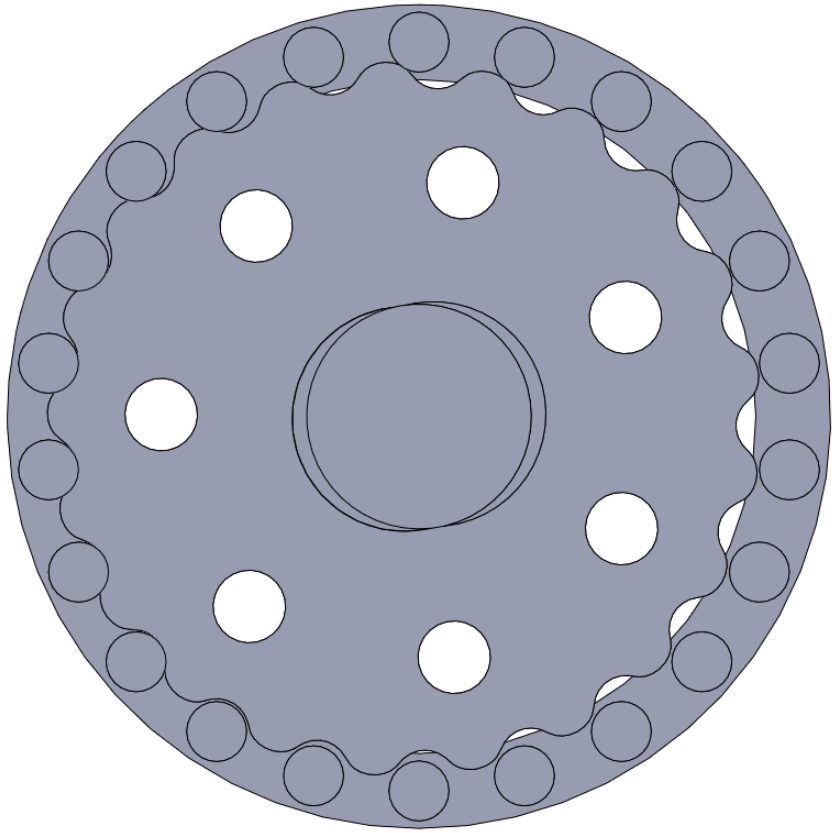
Use cycloidal gear reduction?

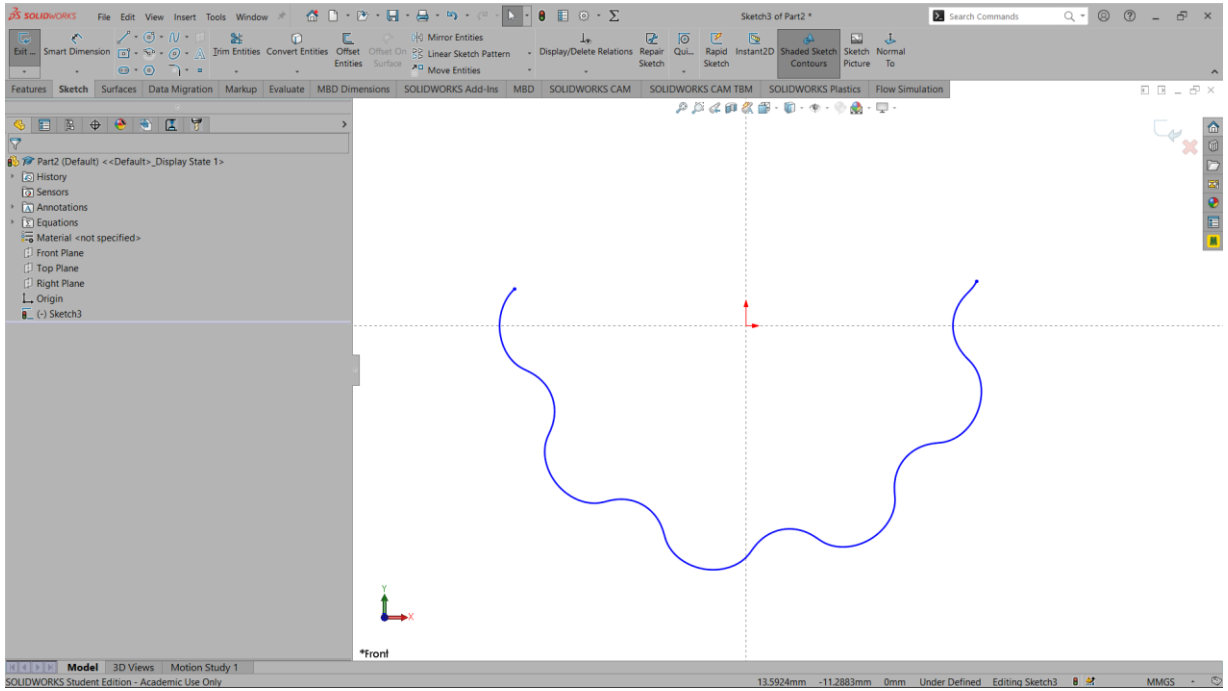


Benefits
- Quiet
- Very compact

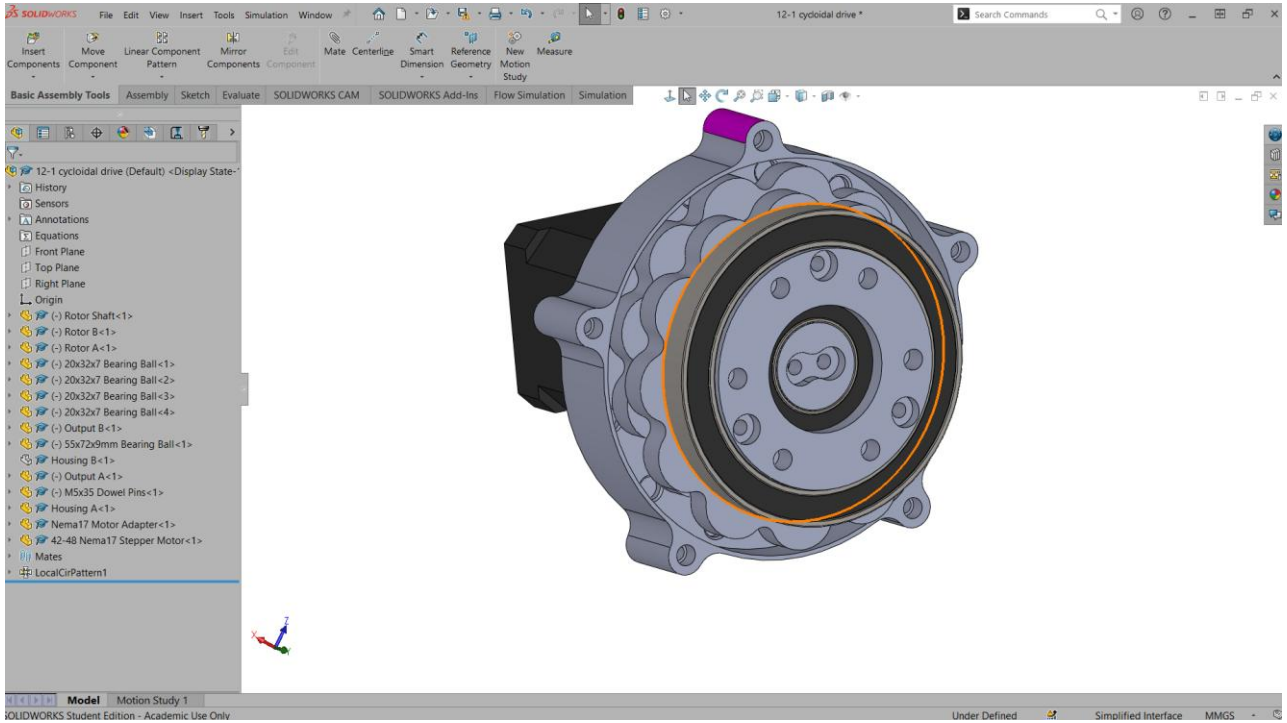
https://www.desmos.com/calculator/neoglqbxig

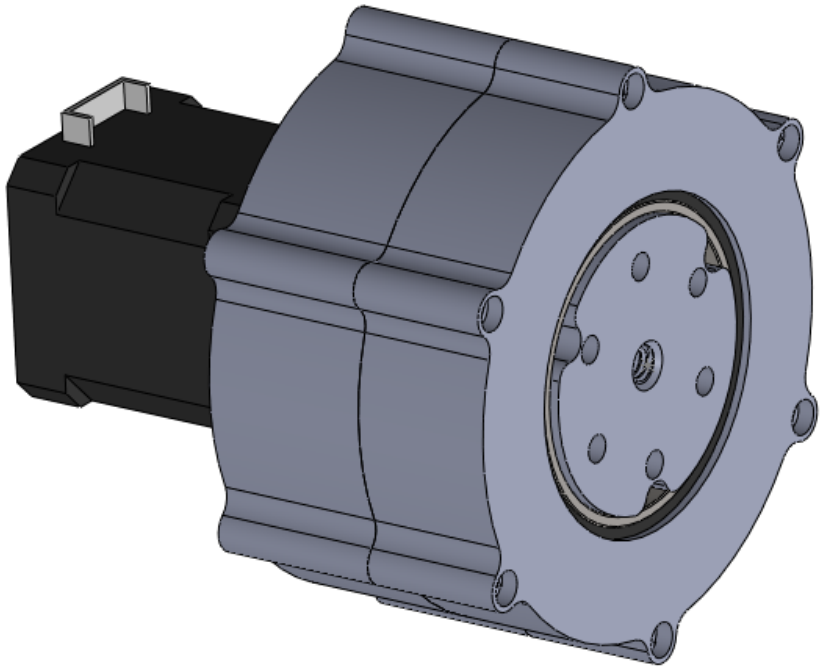
Curve in SolidWorks



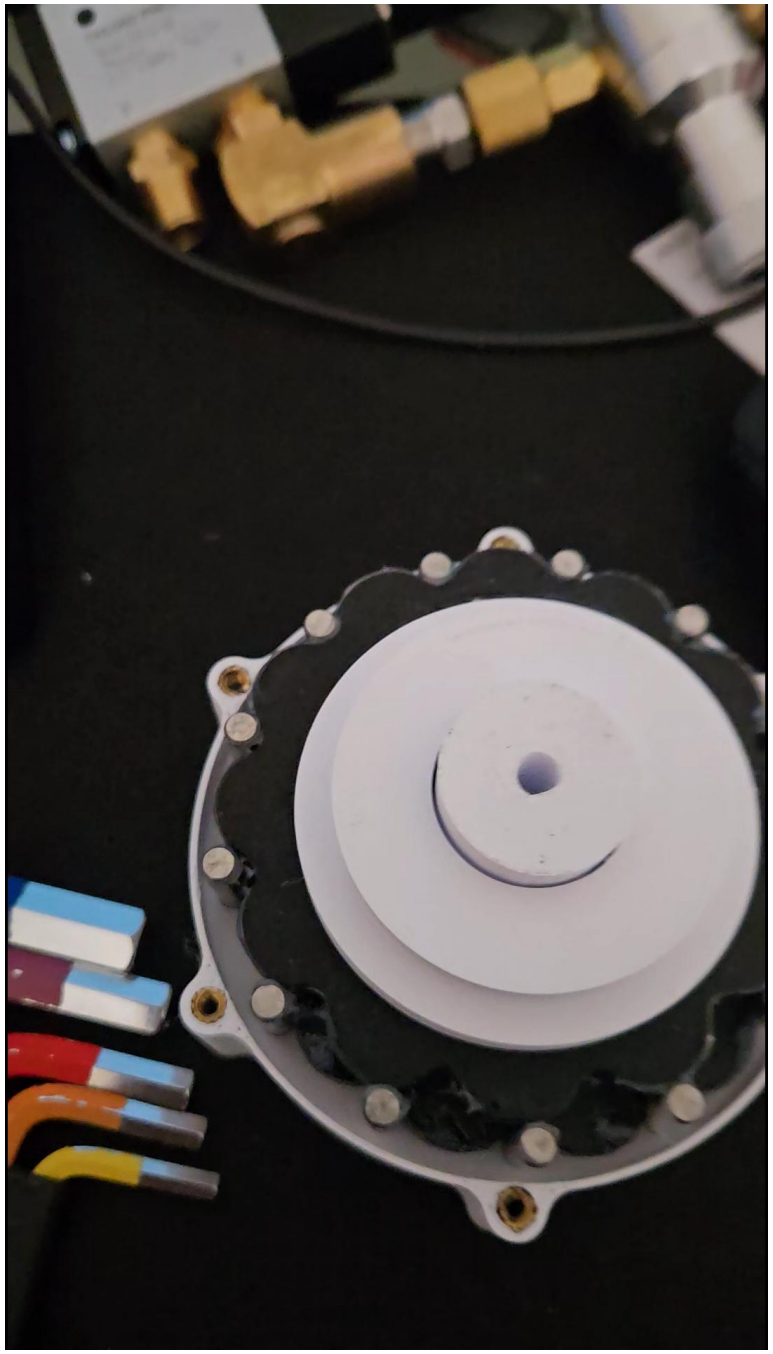


Stepper with reducer

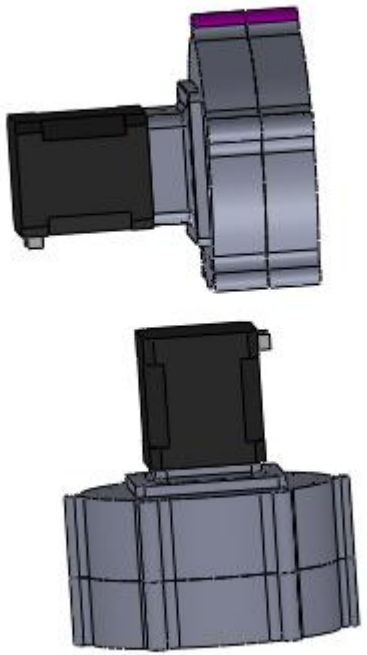




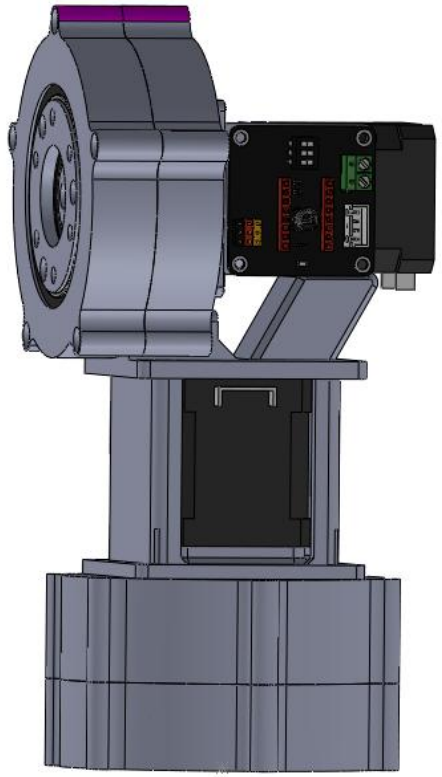
Real life validation



Two perpendiculars sets?



Design all the mating brackets



Designing gas system



3 way solenoid valve

CO_2 cartridge

Regulator

Pressure chamber for cycling shots

Issue

- Solenoid is not cycling with 12 volts.
- Increasing voltage does nothing
- You can hear it click
- People on Amazon say its broken

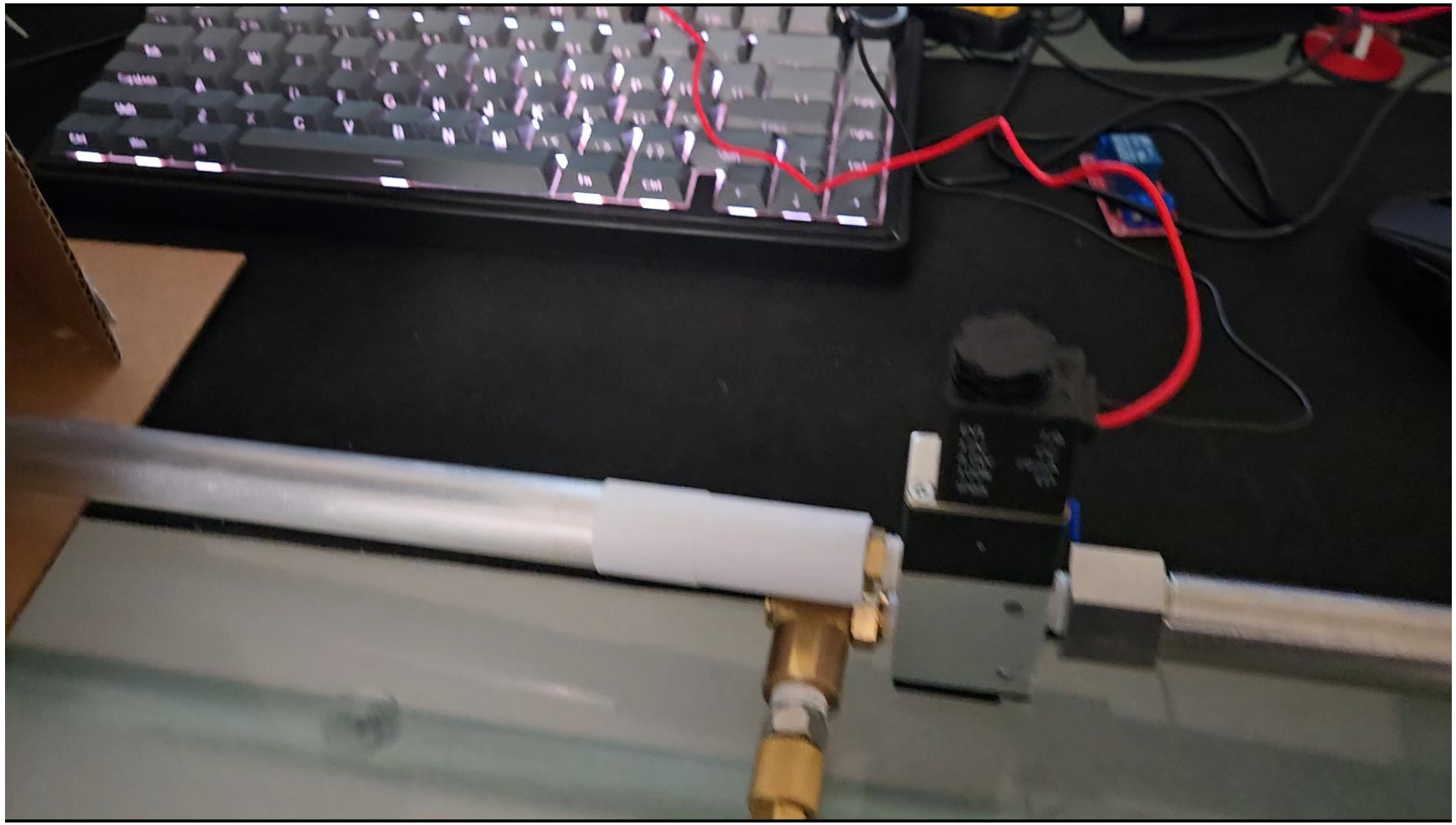
- Manually cycling it works only along one set of valves.

- Oh you need the increase the pressure going to unstick the valve. It doesn’t work below 20 psi.

Figuring out why the solenoid didn’t work



Quick firing test

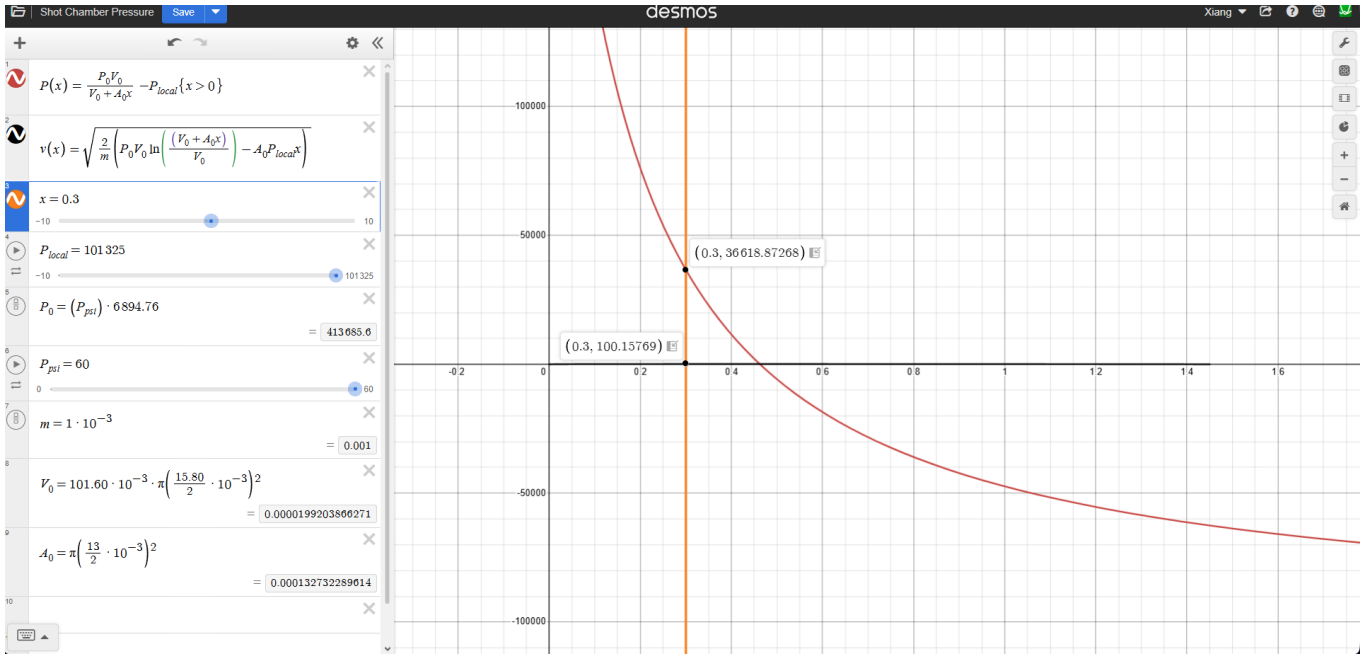


Scary!



It broke the dart

Optimizing Pressure and Length



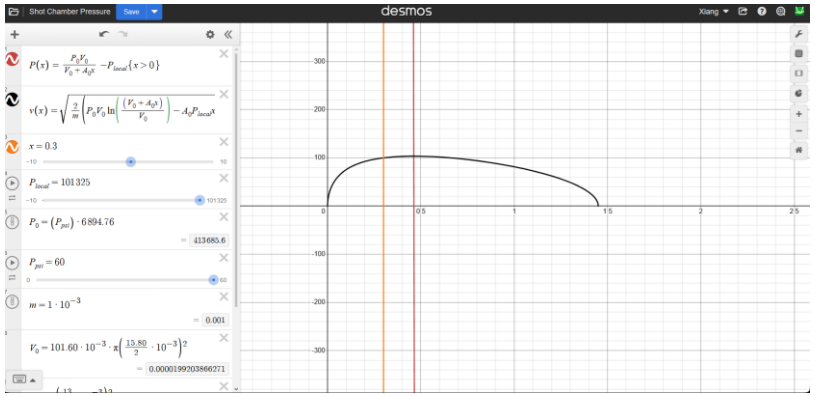
Example
at 60 psi

Legend

Orange is when the dart leaves the chamber.

Red is the pressure in the chamber as the dart travels down it

Black is the velocity of the dart



Different Scale

2 optimization strategy

Optimize length so there is enough pressure left to cycle the chamber

Optimize pressure to make the dart fly out without wasting extras gas

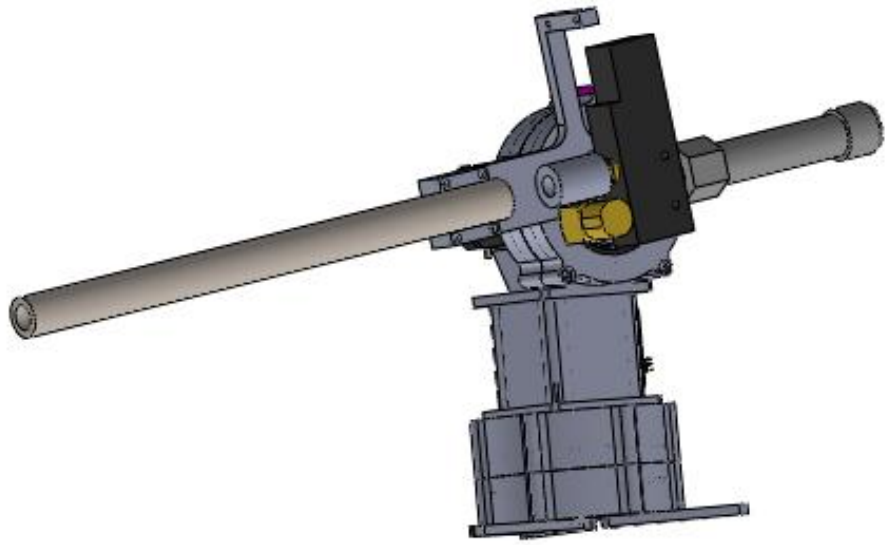
https://www.desmos.com/calculator/b97u9y0dfa

Optimizing Pressure and Length

Caution with the previous calculation.

It does not account for friction so add a little bit extra pressure to overcome it.

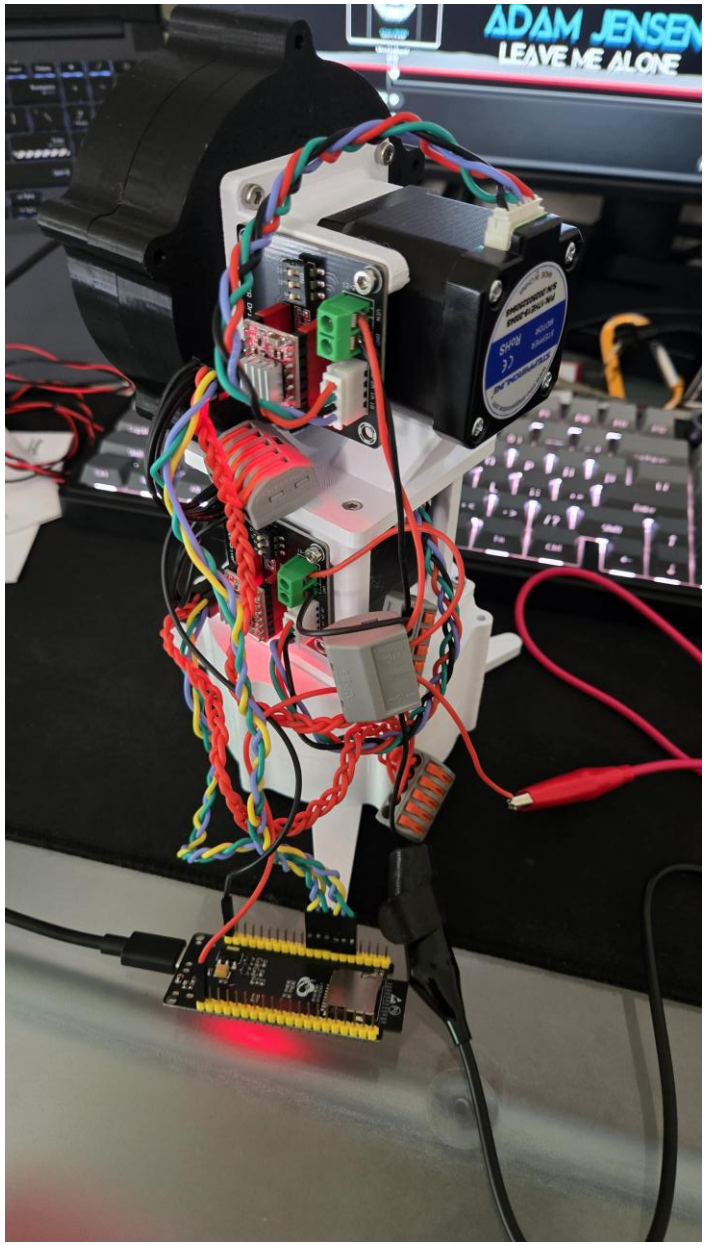
Putting it into CAD



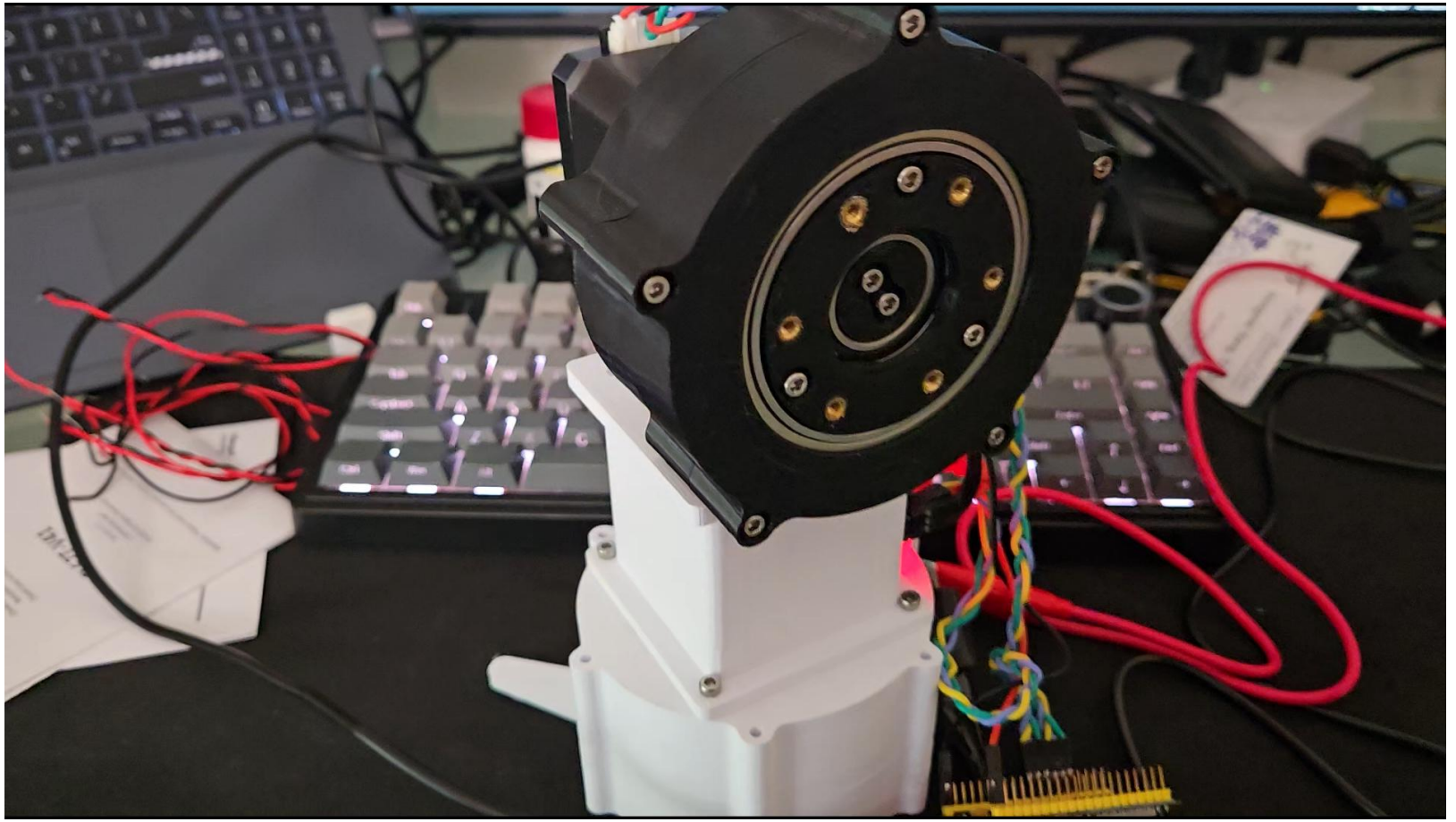
Figuring out ESP32 pins

Pin name	Pin number	HIGH Signal?	Voltage	Safe for GPIO	Notes
G14	14	☑	3.3	SAFE	
G13	13	☑	3.3	SAFE	
G12	12	☑	3.3	SAFE	
G11	11	☑	3.3	SAFE	
G10	10	☑	3.3	SAFE	
G9	9	☑	3.3	SAFE	
G46	46	☐	0	UNSAFE	Bootstrapping (strapping pin)
G3	3	☐	0	UNSAFE	UART0 RX (programming)
G8	8	☑	3.3	SAFE	
G18	18	☑	3.3	SAFE	
G17	17	☑	3.3	SAFE	
G16	16	☑	3.3	SAFE	
G15	15	☑	3.3	SAFE	
G7	7	☑	3.3	SAFE	
G6	6	☑	3.3	SAFE	
G5	5	☑	3.3	SAFE	
G4	4	☑	3.3	SAFE	
G1	1	☐	0	UNSAFE	UART0 TX (programming)
G2	2	☐	0	SAFE	
G42	42	☑	3.3	SAFE	
G41	41	☑	3.3	SAFE	
G40	40	☑	3.3	SAFE	
G39	39	☑	3.3	SAFE	
G38	38	☑	3.3	SAFE	
G37	37	☑	3.3	SAFE	
G36	36	☑	3.3	SAFE	
G35	35	☑	3.3	SAFE	
G0	0	☐	4.3	UNSAFE	Bootstrapping (strapping pin)
G45	45	☑	3.3	SAFE	
G48	48	☑	3.3	SAFE	
G47	47	☑	3.3	SAFE	
G21	21	☑	3.3	SAFE	
G20	20	☐	3.3	UNSAFE	USB D+
G19	19	☐	0	UNSAFE	USB D-

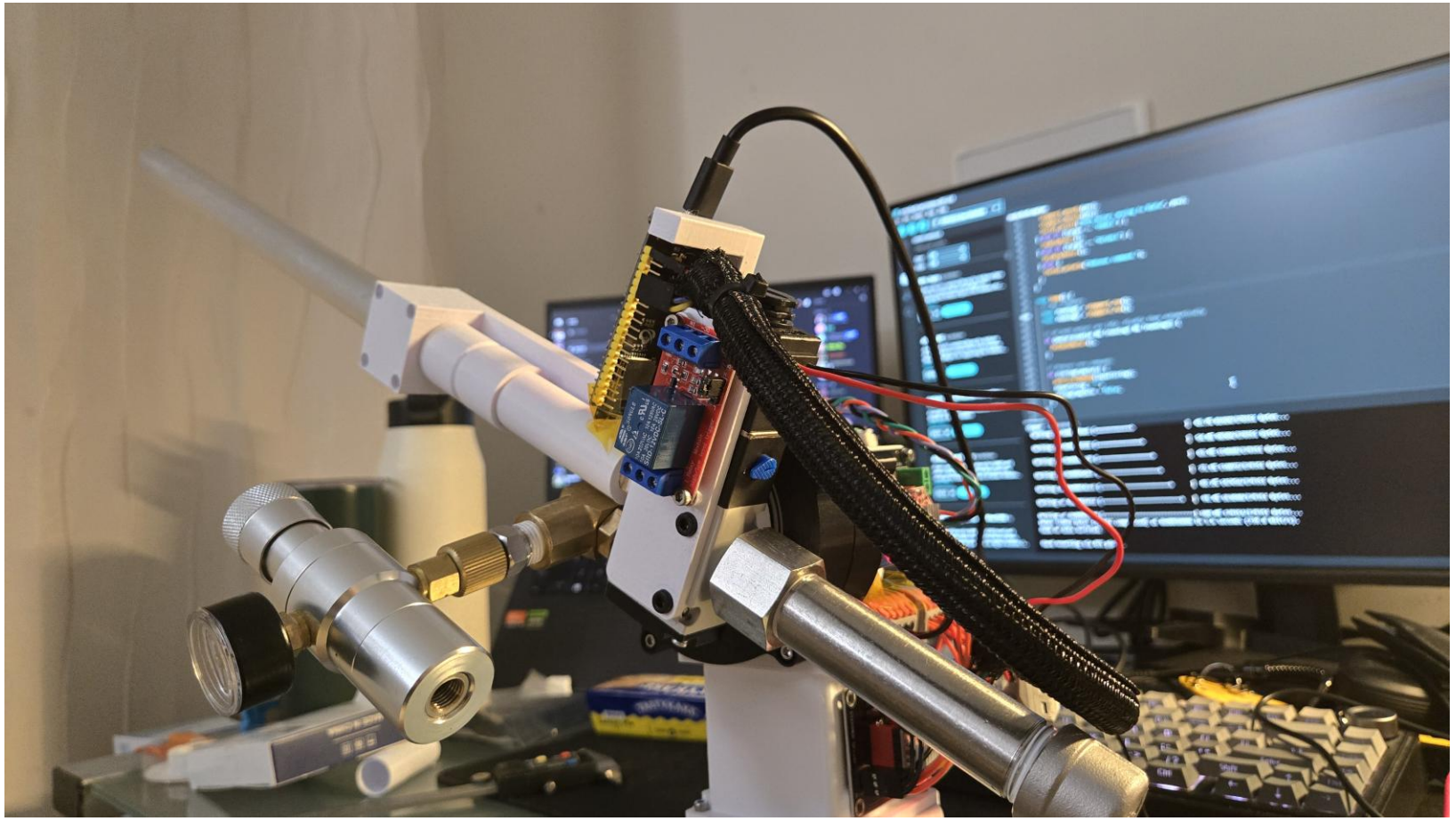
Doing the wiring



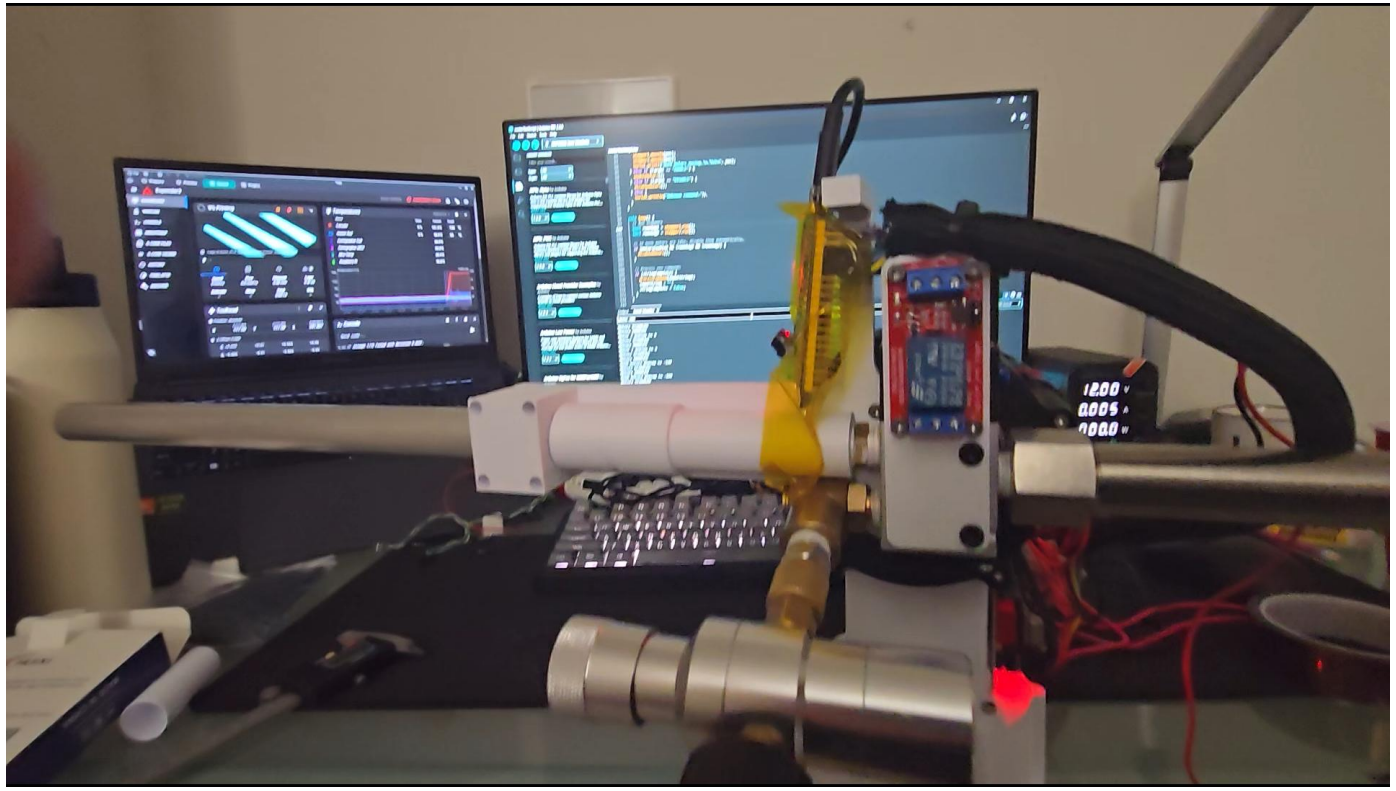
Look at it go!



More wiring

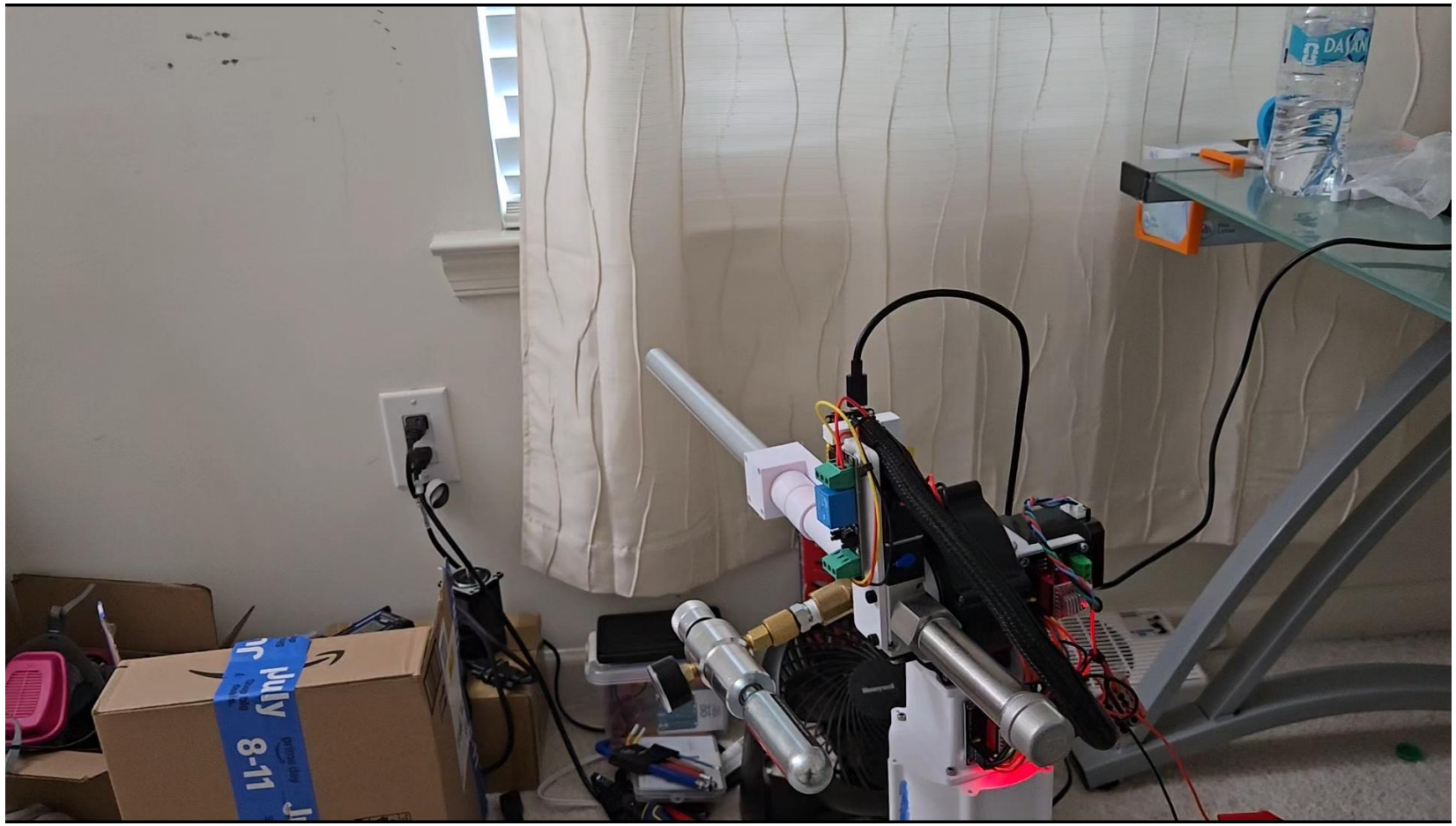


Quick Motion Test



Choppy sound from bad motor drive connection

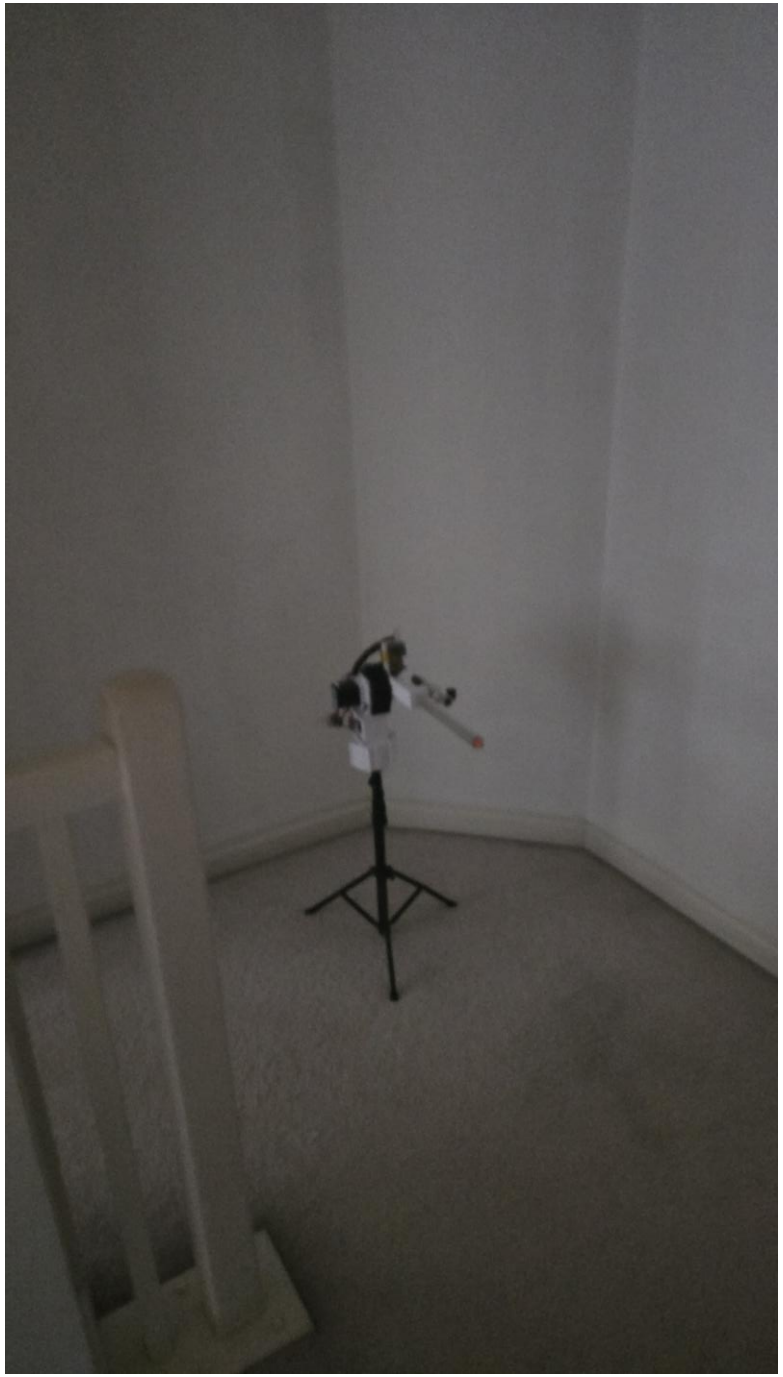
Motion and fire sequence



Gas turned off to save CO2 cartridges.

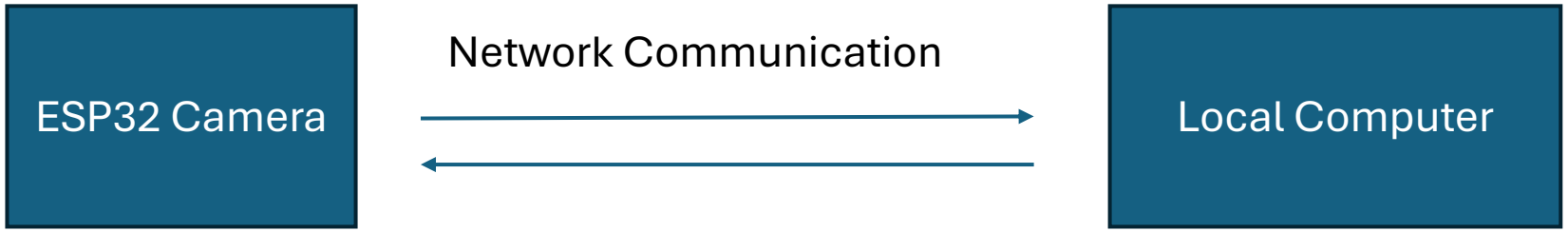
See solenoid LED light up

Turret in my hallway!



Computer Vision and Tracking Work

- Worked in collaboration with Jinson Jimmy (UMD Student)
- Developed a system capable of identifying human features



- Handles Motion Control
- Handles Firing Sequence
- Handles Pictures

- Handles identifying human feature from camera
- Handles calculating motion trajectory

Issues

- The ESP32 was very slow at getting the image and sending it over network
- ESP32 had issues with running facial recognition locally
- ESP32 camera resolution wasn’t the best

End of Project

- Future works
  - Add a battery so the system could be independent of a power supply.
  - Possibly acquire a faster microcontroller or even upgrade to an SBC to do local computation
  - Add automatic dart cycling